After the call with Jérôme

The mathematics, the implementation, the results, and what is left

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Where the eleven questions stand

#	Question	Evidence	Status
Q1	Is Gaussian-message BP provably the same as the linear score? Via Kalman?	derivation	**Done**
Q2	With a Laplace ansatz and a Laplace innovation, is BP exact?	E22, E22c	**Done — no**
Q3	What do the messages look like? Gaussian $* P_\varepsilon$?	E22, E22c	**Done — nearly**
Q4	Closure only for the Gaussian, or any finite family?	E22	**Done**
Q5	Can BP be done in Fourier space?	E23 + analysis	**Done — no**
Q6	Can K be a neural network fitted by EM through BP?	E25	**Open** (3rd attempt running)
Q7	Posterior mean: how long for a network, how long for BP?	E24, E24gpu2	**Done — BP slower**
Q8	Generate without a score: flat marginal, run BP once?	E26, E27	**Done — yes**
Q9	Is the score compared at fixed t or averaged?	E28	**Running**
Q10	Information about (ρ, β) or about (κ_2, κ_4)?	recomputation	**Done**
Q11	How does EM–BP scale in M and L?	E24	**Done — linear**

Eight of eleven answered. Six of them came out *against* what one of us asserted in the call, and three against a prediction of my own that had been written down first. Those are marked throughout.

1 The mathematics

1.1 Q1. Gaussian closure, the LMMSE estimator and the Kalman smoother

Moment-projected sum-product — apply P_G (match mean and variance) to every message immediately after each transition integral — returns the LMMSE estimator of the covariance-matched Gaussian twin,

$$\hat{a}_\mathrm{LMMSE} = \mathrm{Cov}(a, x)\,\mathrm{Cov}(x)^{-1} x = e^{-t}\Sigma_0\big(e^{-2t}\Sigma_0 + \Delta_t I\big)^{-1} x,$$

and Jérôme's guess that "the route is through the Kalman filter" is right: this is exactly the Rauch–Tung–Striebel smoother of the linear-Gaussian state-space model matched to the chain's first two moments. Verified against an RTS smoother written from scratch: $\max|\mathrm{BP} - \mathrm{RTS}| = 6.66 \times 10^{-16}$ on the posterior means, 4.44×10^{-16} on the variances, 2.0×10^{-15} against a dense LMMSE solve, and 8.9×10^{-15} against $-\Sigma_t^{-1} x$ as the score.

The lemma I drafted for this was false, and the way it was false is the point. It dropped the hypothesis "*and an initial message Gaussian or projected by* P_G". That is not a technicality: P_G *does not commute with the likelihood tilt*, so a non-Gaussian initial message is not equivalent to its projection and the error propagates down the chain instead of being absorbed at the first site. Counterexample computed at $L = 32$, $\rho = 0.85$, $q = 0.2775$, $v_0 = 1$, innovation Laplace$(0, 1/\sqrt{2})$, 2×10^5-point quadrature, $t = 1$: the posterior means differ by 4.148×10^{-2} at site 0.

Three further errors in the same draft: $\Sigma_0 = \rho^{|i-j|}$ is missing the factor v_0; the $\rho = 0$ reduction is $\alpha v_i/(\alpha^2 v_i + \Delta_t)\, x_i$, not $\alpha/(\alpha^2 + \Delta_t)\, x_i$; and "both algorithms advance a state in $\mathbb{R}^2$ by the same map" is false for the backward sweep — BP's backward map is autonomous in (λ, r) while RTS's gain $J_i = \rho P_{i|i}/P_{i+1|i}$ depends on stored forward quantities. They agree on the *answer*, not on the recursion. **No new lemma is needed**: the identification is already in `reports/compendium/parts/12-gaussian-closure-lmmse.tex` and in the archival thesis at 80 configurations with max disagreement 1.3×10^{-15}. One sentence in Proposition 1's remark closes it.

1.2 Q5. The BP message in Fourier, in closed form

Write $K(a'|a) = p_\varepsilon(a' - \rho a)$ and $g_i = L_i \ell_i$ for the tilted message. One forward step is $L_{i+1}(a') = \int p_\varepsilon(a' - \rho a) g_i(a)\, da$, and substituting $y = \rho a$ turns it into a convolution with a *dilated* g_i. Hence

$$\widehat{L}_{i+1}(\omega) = \varphi_\varepsilon(\omega)\, \widehat{g_i}(\rho\omega). \tag{1}$$

Two of the three BP operations are free in Fourier: the transition is the dilation $\omega \mapsto \rho\omega$, and adding the innovation is the multiplication by φ_ε. With a flat likelihood $\widehat{g_i} = \widehat{L_i}$, and unrolling (1) from $\widehat{L_1} = \varphi_\mu$,

$$\boxed{\widehat{L}_{k+1}(\omega) \;=\; \varphi_\mu(\rho^k\omega)\prod_{j=0}^{k-1}\varphi_\varepsilon(\rho^j\omega).} \tag{2}$$

Verified against this project's own `grid_bp`: max absolute error 2.16×10^{-8} at sites $1, 2, 4, 8, 16, 31$, which is the grid's quadrature error at $h = 0.006$ and not model error.

As $k \to \infty$, $\varphi_\mu(\rho^k\omega) \to 1$ and the message tends to $\prod_{j\geq 0}\varphi_\varepsilon(\rho^j\omega)$ — the stationary law's characteristic function, **an infinite product**. This explains a prediction of ours that E22 falsified: we expected the Gaussian-ansatz error to be *smallest* at large t and it *grows*, because with a near-flat likelihood the recursion drives messages to exactly this object, which lies in no finite-parameter family.

The decay law. For Laplace(b), $\varphi_\varepsilon(\omega) = (1 + b^2\omega^2)^{-1}$, so $\log|\widehat{L}_\infty| = -\sum_j \log(1 + b^2\rho^{2j}\omega^2)$. The terms with $b\rho^j|\omega| \gg 1$ contribute $-2\log(b\rho^j|\omega|)$ and there are $J = \log(b|\omega|)/\log(1/\rho)$ of them; summing the progression gives

$$\log\bigl|\widehat{L}_\infty(\omega)\bigr| \;\sim\; -\frac{\bigl(\log b|\omega|\bigr)^2}{\log(1/\rho)}. \tag{3}$$

A Gaussian decay in $\log\omega$ — super-polynomial, sub-exponential. Fitted over $3 \leq \omega \leq 3 \times 10^4$:

ρ	0.50	0.70	0.85	0.95	0.99
fitted slope	−1.5339	−2.8988	−6.2578	−19.654	−100.55
$-1/\log(1/\rho)$	−1.4427	−2.8037	−6.1531	−19.496	−99.499
ratio	1.063	1.034	**1.017**	1.008	1.011
R^2	0.99993	0.99995	**0.99997**	0.99999	0.99999

The ratio tends to 1 as $\rho \to 1$, as it must: the derivation assumes many terms in the product. **Do not use (3) numerically at practical tolerances** — inverting it gives $\omega_{\max}(\eta) \approx b^{-1}\exp\sqrt{\log(1/\eta)\log(1/\rho)}$, which over-predicts by 1.4–1.8× at $\eta = 10^{-2}\ldots10^{-6}$ (6.4/9.1/12.0 predicted against 3.6/5.9/8.3 measured), because the fitted intercept of −6.57 nats is the whole budget when $\omega_{\max}$ is only 4–8. Evaluate the product in (2) instead; it costs nothing. What matters is that **the truncation level is analytically computable**.

The likelihood tilt is the one hard operation, and it is hardest at *low* noise. $\ell_i \propto \mathcal{N}(a; x_i/\alpha, \Delta_t/\alpha^2)$ is a product in real space, hence a *convolution* in mode space with a Gaussian of width

$$\sigma_\omega = \frac{\alpha}{\sqrt{\Delta_t}} = \frac{e^{-t}}{\sqrt{1 - e^{-2t}}},$$

which is *decreasing* in t. On a domain of half-width W the mode spacing is π/W, so the bandwidth is $\approx 3\sigma_\omega W/\pi$ ($W = 6$):

t	0.05	0.073	0.105	0.153	0.222	0.467	0.982	2.07	3.00
σ_ω	3.08	2.53	2.07	1.67	1.34	0.81	0.40	0.13	0.05
bandwidth (modes)	**17.7**	**14.5**	11.8	9.6	7.7	4.6	2.3	0.7	**0.29**

Banded at ten of twelve levels, effectively dense at the two lowest, and **the hard end is low noise** — the opposite of the natural guess. At $t = 3$ the kernel is 0.29 modes wide, so the tilt is free. This answers the tractability question and does *not* revive the solver; see §2.2.

1.3 Q5. Why the cumulant basis fails where Fourier does not

The Hermite / Gram–Charlier expansion converges in L^2 only under **Cramér's condition** $\int q(z)^2/\phi(z)\,dz < \infty$, with q the density being expanded and ϕ the standard normal. It **fails for every standardised BP message, at every noise level**:

t	msg. var	$A = 8$	$A = 12$	$A = 16$	$A = 20$	growth
0.05	0.331	2.3e23	5.5e66	1.7e131	5.6e216	×2.4e193
0.222	0.441	2.7e13	4.9e43	6.3e89	5.1e151	×1.9e138
0.982	0.751	60.5	1.9e15	1.6e38	2.8e70	×4.6e68
3.00	0.992	1.09	4.9e6	4.7e21	5.3e43	×4.9e43

A Fourier basis is complete in L^2 for any finite-variance density. **One basis has a convergence theorem this chain satisfies and the other has one it violates** — that is the structural reason for the measurement in §2.2, and it replaces the weaker argument we had been using (that $|\kappa_6|$ runs 1–228, a symptom rather than a cause).

Getting here required refuting two explanations of my own, and both refutations are why the third can be trusted. *(i)* "It is the metric and the positive cone" — refuted: the *unweighted* L^2 error of the truncation diverges too, and starts diverging *before* any negative mass appears (at $t = 0.2216$ the order-6 truncation has zero negative mass and the L^2 error has already risen from 2.17e-2 to 7.58e-2). *(ii)* "It is my arithmetic" — refuted by a control with a known answer: for a Gaussian message every c_n with $n \geq 3$ must vanish, and the measurement gives $\max_{n\geq 3} |c_n| = 1.9 \times 10^{-18}$ with order-30 reconstruction error 2.9×10^{-13}, on three grids.

And my first Cramér test was itself wrong twice over, which is the methodological lesson of the whole section. It evaluated the integral on the *raw* message (variance 0.33–0.99, narrow, so it converged) rather than the *standardised* one the expansion is applied to; and it hunted the divergence by *refining the grid*, which reads 6114.3, 6113.3, 6113.1, 6113.0 at $N_g = 1601 \ldots 12801$ on a fixed domain and detects nothing. **Grid refinement tests quadrature; only widening the domain tests tail integrability.**

1.4 Q8. One BP sweep is an exact sampler

Jérôme: "*the exact BP, so not even a score — I think we don't even need it to generate. We can just start from a flat marginal and run BP once.*" On a chain this is exactly right, and sharper than it sounds. Sum-product returns the exact conditional laws, so forward-filtering–backward-sampling draws from the chain posterior in one sweep: filter $F_i(a) \propto \ell_i(a) \int K(a|a')F_{i-1}(a')da'$, then draw $a_L \sim F_L$ and, backwards, $a_i \sim F_i(a_i)K(a_{i+1}|a_i)$. No score, no reverse SDE, no discretisation in t.

With a *flat* likelihood the backward message of a normalised kernel is constant, so sum-product degenerates to plain ancestral sampling $a_1 \sim \mu$, $a_{i+1} \sim K(\cdot|a_i)$ — the "flat marginal" limit. That constancy holds only up to grid error, and the two grid errors behave differently, which is worth recording: the kernel's column mass is $1 + O(h^2)$ (3.8e-3, 9.6e-4, 2.4e-4, 6.0e-5 at $N_g = 201, 401, 801, 1601$) while the *truncation* error saturates at 2.1e-2 and never improves with N_g, because at $|a| \sim A = 8$ the kernel loses mass off the edge of the grid. That region lies beyond five standard deviations and carries no prior mass, but refining N_g will never remove it.

1.5 Q10. Information about the cumulants is a recomputation, not a relabel

Jérôme: "*to resolve β you need all the cumulants ... it is more the information of the second cumulant and of the fourth.*" He is right, and the fix is a change of coordinates, so it needs the delta method. For the generalised-Gaussian kernel, which holds $\mathrm{Var}(\varepsilon) = q$ at every β,

$$\kappa_2 = q, \qquad \kappa_4 = \gamma_2(\beta)\, q^2, \qquad \gamma_2(\beta) = \frac{\Gamma(5/\beta)\Gamma(1/\beta)}{\Gamma(3/\beta)^2} - 3,$$

so with $\theta = (\rho, q, \beta)$ and $\varphi = (\rho, \kappa_2, \kappa_4)$ the Jacobian $J = \partial\varphi/\partial\theta$ is lower triangular with rows $(1, 0, 0)$, $(0, 1, 0)$, $(0,\ 2\gamma_2 q,\ \gamma_2' q^2)$, and

$$I_\varphi^{-1} = J\, I_\theta^{-1} J^\top, \qquad \mathrm{eff}(\varphi_k) = 1/\big(j_k^\top I_\theta^{-1} j_k\big), \quad j_k = \text{row } k \text{ of } J.$$

ρ and κ_2 come out unchanged because their rows are unit vectors — which is the internal check that J is assembled correctly — and κ_4 does not, because j_3 mixes in the q direction. **A relabel would have been wrong by a factor between 2.3 and 1518.**

2 Results

2.1 Q2–Q4. The message shapes, confirmed at scale (E22, E22c)

Jérôme's ansatz — a Gaussian pushed through *one* transition of the true chain — is exact at site 1 (10^{-15}, at the measured quadrature floor) and fails by $\sim 10^{10}$ by site 31. It nonetheless beats a plain Gaussian by 1600–16000× and a four-parameter mixture by 50×, so the conjecture is very nearly right and the failure is structural rather than a matter of fit.

E22c swept $\rho \in \{0.5, 0.7, 0.85, 0.95\}$, $L \in \{32, 128\}$, $N_g \in \{801, 1601\}$, four innovation families, twelve levels: **both verdicts hold in 432/432 cells**, with growth factor from site 1 to the deepest site spanning 8.0e7 to 1.15e11. **The load-bearing control:** doubling the grid does not move the error — KL(1601)/KL(801) = 1.000 over 1512 cells — so the failure is the *model*, not the quadrature, which is the one alternative that would have made the result vacuous.

2.2 Q5. The Fourier solver is not worth building (E23 + the analysis)

E23 measured the premise before committing to a solver. On the Laplace chain, 144 cells per representation, KL of the reconstructed message against the true one:

representation	dof at 10^{-2}	dof at 10^{-4}	cells reaching 10^{-4}	non-monotone
grid (uniform trapezoid)	45	151	**144/144**	**0/144**
Fourier, shared domain	101	151	140/144	92/144
Fourier, adapted per message	**33**	**47**	53/144	10/144
cumulant (Gram–Charlier)	5	—	24/144	**144/144**

Jérôme's intuition is quantitatively right where it converges: the adapted Fourier basis beats the grid in **53/53** cells where both reach 10^{-4}, median ratio **4.28**× (47 dof against 151), and 47 dof $\approx$ 23 modes — close to his "the first 20 Fourier modes". **It is nevertheless unusable as a solver basis**, because it reaches 10^{-4} in only 53 of 144 cells and is non-monotone in the truncation in 10; a solver must be able to *choose* a truncation and know the error it accepts, and the grid is monotone in 144/144.

Even the 4.28× would not have bought 4.28×: E24's anatomy part measured that only **23.8%** of a BP sweep at $N_g = 401$ is the N_g^2 matmul (the rest is $O(BLN_g)$ elementwise work and traffic over 158 MB of message arrays), so making the spectral step free caps the speed-up at $1/(1 - 0.238) = 1.3\times$. **The one substantive change the analysis made is that the tilt — which we had assumed was the blocker — is not. The blocker is that BP is memory-bound, so there is nothing for a faster spectral step to buy.**

A by-product worth keeping: uniform trapezoid quadrature is not merely adequate but correct here. Gauss–Legendre and Clenshaw–Curtis are $\sim 10\times$ *worse*, confirmed over 144 cells — for a smooth decaying integrand the Euler–Maclaurin terms cancel and the trapezoid rule is spectrally accurate, while the Laplace kink is not polynomial-like.

2.3 Q7 and Q11. Inference cost, on both devices (E24, E24gpu2)

At the frozen operating point ($\rho = 0.85$, $L = 32$, $N_g = 401$, batch 512), milliseconds per posterior mean per sequence:

	bp_grid	transformer	bigru	tcn	unet	window
CPU	0.3196	0.0254	0.0283	0.0148	0.0127	0.0025
H200 GPU	0.0400	0.0028	0.0012	0.0020	0.0024	0.0010

The call's guess that BP should be faster was wrong, and so was our own first correction of it. Exact grid BP costs $\sim 15\times$ more per posterior mean than the transformer on the CPU (12.6×) *and* on the H200 (14.3×). An earlier note in this project recorded that "BP is slower at inference was a CPU artefact"; that was wrong. What is a CPU artefact is the *absolute* BP number, not the ratio.

What the GPU did change is the mechanism. On the H200 `bp_grid`'s wall clock is **flat at 0.0204–0.0215 s from batch 32 to 2048** — a 64× range for a 5.6% change. It is launch-latency-bound, and inherently so: the L-site recursion is serial, so a sweep is $2(L - 1) = 62$ sequential kernel launches however wide the batch. The transformer is flat only to batch 512 and then becomes compute-bound. So the ratio narrows — 15.2×, 14.8×, 10.3×, **6.1**× at batch 32, 512, 1024, 2048 — and extrapolating gives a **crossover near batch $\approx$ 12 600**. The honest framing is neither "BP is slower" nor "BP is faster": *BP pays a fixed serial latency and amortises it over the batch; the network pays per example and saturates.*

Q11. Scaling is **linear** in the number of sequences, not quadratic as answered in the call (slope +1.04 full range, +1.19 top half), and linear in L (+1.12/ + 1.13). Our *own* predeclared N_g exponent of 2.0 was also falsified: measured +1.27/ + 1.47, because the operation count is quadratic while the wall clock is not until $N_g \gtrsim 1600$.

Training, which is where the argument actually lives. The M-step wall clock is **flat in dataset size** — exponent +0.008 (mixture) and −0.037 (MDN) over a 64× range — while the E-step is linear (+0.77 to +0.95). And the exponent stays ≈ 0 at *every* capacity: 136× more kernel parameters costs 1.8× more per M-step against 6.3×-per-step for 175× more transformer parameters. **Stated narrowly, because the totals do not support more:** at $M = 8192$ the E-step (4.1 s) dwarfs the MDN M-step (0.13 s), so EM–BP's total training cost is *comparable* to a transformer's, not orders below. The claim is about parameters, not totals.

2.4 Q8. Generation without a score (E26, E27)

Weak form: does score accuracy transfer to generation? Yes, sharply. E26 drove the same 400-step reverse SDE on identical Brownian paths with four scores:

score	excess kurtosis of the AR residual	reference	KL of the residual histogram
exact (true-kernel BP)	1.66–1.78	1.87	6.2e-4
em_bp (fitted mixture)	1.18	1.87	2.6e-3
lmmse (covariance twin)	−0.01	1.87	2.2e-2
isotropic	−0.03	1.87	3.8e-1

The load-bearing row is lmmse: excess kurtosis -0.01 **where the truth is 1.87.** The covariance-matched Gaussian score gets the covariance right (`cov_fro` 0.069 against the exact score's 0.071) and destroys every higher-order structure in the law. This is the clearest statement in the project of what the "learned scores are basically a low-pass filter of a Gaussian" framing throws away.

Strong form: is the reverse SDE needed at all? No (E27). Four arms, all emitting samples at $t_{\min}$ against E26's own reference stream so the two experiments are paired. Medians over 4 seeds × 2 training sizes:

arm	KL of residual hist.	excess kurtosis	SW_2	W_1
`mc_floor` (i.i.d. from the true law)	**0.00074**	1.884	0.0495	0.0130
`ancestral_true` (one BP sweep)	**0.00074**	1.895	0.0487	0.0139
`ffbs_true` (one sweep, conditioned at $t = 3$)	0.00072	1.914	0.0497	0.0146
`ancestral_em` (fitted kernel)	0.00194	1.831	0.0581	0.0328
reference	—	1.919	—	—

One BP sweep generates at the Monte Carlo floor — statistically indistinguishable from i.i.d. draws from the true law. And conditioning on an observation at $t_{\max} = 3$ ($\alpha = 0.0498$) changes nothing measurable, which is the quantitative content of "flat marginal". The grid-quantisation control, paired on identical grid indices, moves the KL by 5.8×10^{-5}: negligible.

Cost at equal error, which is the honest comparison.

n_{steps}	25	50	100	200	400	800
KL	0.00340	0.00146	0.00088	**0.00069**	0.00075	0.00077
KL / one sweep	4.58	1.96	1.19	0.93	1.01	1.04
ms / seq (batch 2048)	13.2	26.1	54.7	**116.9**	231.5	459.0

One sweep costs **0.161 ms** per sequence at batch 2048. The reverse SDE first matches it at ≈200 steps and then saturates — so at equal error one sweep is **725× cheaper**, and against the 400-step resolution E26 used, **1436×**.

This falsifies a prediction of mine and corrects E26. I predicted one exact sweep would *beat* the 400-step reverse SDE on kurtosis. It does not: the 400-step SDE is also at the Monte Carlo floor. So E26's apparent deficit for the exact score (1.66 against 1.87) was Monte Carlo scatter, not discretisation bias. The `mc_floor` arm existed precisely to catch that, and E26 had no such arm — which is why its number could not be read.

What this does to the framing. The diffusion apparatus here is not needed to *sample*; it is needed to *learn the kernel from noisy observations*. Once the kernel is known, sampling on a chain is one sweep. That is Jérôme's point, and it is a statement about what the paper is about rather than only about cost.

2.5 Q6. The neural kernel — still open, and why (E25)

This is Jérôme's headline suggestion and the one question still without an answer. Two attempts were void for unrelated reasons; a third is running. The landed second attempt shows what is at stake (median test risk over 16 seeds):

M	32	128	512	2048
`em_bp` (8-component mixture, 25 params)	0.0931	0.0700	0.0398	**0.0229**
`mdn` (small network, 1516 params)	0.3836	0.2732	0.3800	0.3678

The MDN row is not a model result. Its selected checkpoint is 1, 3, 4 or 5 in every one of 64 cells, with `converged=True` in 64/64: the arm stopped after ∼10 EM iterations of a 400-iteration budget. Reported naively that reads "the neural kernel is 10× worse". It is an arm that never trained. Three defects, all in shared machinery rather than in the driver: `fit_em` declared convergence on *one* below-tolerance increment (measured increments

$0.130, 0.032, 0.012, 0.018, 0.007, \ldots$ — plainly still climbing — then stop at iteration 10); `MDNKernel.m_step` mutated its receiver and returned `self`, so four "checkpoints" were four references to one object and validation selection was a no-op, disabling the exact overfitting control the experiment exists to test; and the stall was permanent because the learning rate was never reduced on a rejected step. After all three: 121 iterations, 6/6 distinct checkpoints, monotone evidence.

Also note `em_bp` is `censored` in 64/64 cells — it never converges inside 1200 iterations, so its numbers are a budget-limited lower bound on its own performance.

Predeclared reading of the third attempt, written before the data lands. The two arms do not get matched iteration budgets (1200 against 400), and the asymmetry favours `em_bp`. That is conservative for the claim "a small neural kernel helps": if the MDN still wins it wins against a better-resourced baseline; if `em_bp` wins, the result is confounded with budget and must be reported as "`em_bp` at least matches, and had not finished improving". A matched-budget rerun is queued either way.

2.6 The parameter-efficiency claim, tested where it can break (E29)

The claim "a small network for the kernel instead of a huge one for the score" compares 1516 kernel parameters against 135 825 in a transformer, and the first question a referee asks is what a *1516-parameter transformer* does. E29 is a capacity ladder per architecture, designed so a single counterexample kills the claim. The transformer ladder has landed (12 seeds); the entry is the ratio to `em_bp` at the same M, so > 1 means the network is worse:

transformer params	$M = 32$	128	512	2048
843	2.24×	2.01×	3.14×	**5.46**×
2 579	2.35×	2.11×	3.12×	5.40×
5 193	2.54×	2.17×	2.41×	2.63×
18 057	2.64×	2.26×	2.42×	2.76×
135 825	**2.72**×	2.37×	2.69×	3.12×

No transformer at any capacity matches `em_bp`'s 25-parameter mixture kernel, at any training size. **And a prediction of mine is falsified**: I predicted risk would fall monotonically with parameter count. At $M = 32$ and 128 it *rises* — 2.24× to 2.72× — so more capacity is actively worse when data is scarce. That is overfitting, and it is the sample-efficiency claim appearing in the network arm itself. Five of six architectures are still running.

2.7 Q10. Information about the cumulants (recomputation)

Across β at fixed noise level $t = 0.4665$, medians over 8 replicates, $q = \kappa_2 = 0.2775$:

β	γ_2	γ_2'	$\mathrm{eff}(\beta)$	$\mathrm{eff}(\kappa_4)$	$\mathrm{eff}(\kappa_4)/\mathrm{eff}(\kappa_2)$
1.00 (Laplace)	3	−8.50	0.1382	0.288	0.033
1.25	1.527	−4.05	0.0374	0.379	0.045
1.50	0.762	−2.30	0.0140	0.443	0.050
2.00 (Gaussian)	0	−1.00	0.00284	0.479	0.057
3.00	−0.582	−0.333	**0.000343**	**0.525**	0.061

The qualitative reading of the β curve inverts. $\mathrm{eff}(\beta)$ falls by 403× from $\beta = 1$ to 3 (48.5× at $t = 0.05$, 440× at $t = 1.425$), which reads naturally as "shape information collapses as the law approaches the Gaussian". It does not: $\mathrm{eff}(\kappa_4)$ *rises*, by 1.83×. The collapse is the coordinate going degenerate — $|\gamma_2'|$ shrinks from 8.50 to 0.333, a factor 25.5, and it enters $\mathrm{eff}(\kappa_4)$ *squared*, so $25.5^2 = 650$ accounts for the observed 403. Near $\beta = 2$ a change in β barely moves any cumulant, so β is nearly unidentifiable *however much data you have*. That is exactly Jérôme's objection, now a measured factor.

The decay-in-t claim survives and is stronger in the honest coordinates, which was the thing at risk. Over t from 0.05 to 4.0 on the Laplace chain: $\mathrm{eff}(\rho)$ decays 9.1×10^5, $\mathrm{eff}(\kappa_2)$ 3.6×10^6, $\mathrm{eff}(\kappa_4)$ 5.9×10^8 — so **κ_4's information decays 163× faster than κ_2's**, widening to 1.6×10^6 at $\beta = 3$. The parameterisation-free form of the project's central claim is the ratio $\mathrm{eff}(\kappa_4)/\mathrm{eff}(\kappa_2)$, which runs $0.071 \to 0.033 \to 0.0048$ at $t = 0.05, 0.47, 1.43$: shape carries 7% of scale's information at low noise and 0.5% by $t = 1.43$.

Verified rather than assumed: γ_2' in closed form against a central difference at seven β; the reparameterisation inverts to 10^{-9}; and I_θ, which has to be reconstructed from the stored `diag_*/corr_*` because the matrix itself was not saved, is *checked against the stored* `eff_*` before anything is computed from it (worst relative error 8.6×10^{-13} over 560 rows), with a non-zero exit if it fails.

3 Implementation

3.1 What was built

`experiments/e22_message_shapes.py`	Jérôme's ansatz against the true messages; the candidate is built from the recursion's own kernel matrix, not from a convolution lookup.
`experiments/e23_representation.py`	dof per representation, plus the quadrature-rule comparison; `fourier_adapted` added after realising a shared-domain-only test was a strawman.
`experiments/e24_inference_cost.py`	five parts: `cost`, `scaling`, `anatomy`, `training`, `capacity`; `E24_REQUIRE_GPU=1` fails closed.
`experiments/e27_direct_sampler.py`	the one-sweep samplers, batched FFBS, the Monte Carlo floor arm and the paired jitter control.
`experiments/e28_per_level_ranking.py`	per-level risk for every arm, with a `verify` part that refuses the output unless it pools back to the registered estimand.
`experiments/e29_matched_parameters.py`	the capacity ladder; imports E03's trainer and bundles rather than reimplementing them.
`tools/cumulant_information.py`	the delta-method reparameterisation, with the reconstruction check.
`tools/fourier_analysis.py`	the three analytical results, each verified against `grid_bp` or against a control with a known answer.

Changes to the library: `grid_bp` gained `return_messages=True` (exposing normalised forward/backward messages, which E22 and the FFBS sampler both need); `fit_em` gained `patience`; `MDNKernel.m_step` now returns a new kernel instead of mutating its receiver; `backend.gpu_available` retries and records the probe error instead of swallowing it.

3.2 Verification

New tests, all of which pin a derivation against the project's own exact solver or against a case with a known answer rather than against a previously recorded number: `test_ffbs_sampler.py` (8), `test_fourier_analysis.py` (11), `test_cumulant_information.py` (10), `test_em_patience.py` (3), `test_kernel_mstep_contract.py` (6), `test_no_degenerate_bundles.py` (1, an AST walk with no allowlist). The FFBS sampler is checked three ways: its forward filter equals the normalised product of `grid_bp`'s forward message with the local likelihood, to 1.4×10^{-15}; the last site's filter equals the last site's BP belief to 8.9×10^{-16}; and 20 000 *draws* of one fixed observation reproduce `grid_bp`'s posterior means within four Monte Carlo standard errors at every site, with variance matching $\mathrm{Var}_{\mathrm{BP}} + h^2/12$.

3.3 Discipline that paid for itself

guard	what it caught
`test_no_degenerate_bundles`	a *fifth* instance of the re-seeded-comprehension defect, in E27's own `mc_floor` control arm, *before* anything ran. Written after E25 was voided by the fourth.
E28's `verify` part	that E28's per-level decomposition was of a different estimand — correctly, and for a reason I got wrong twice before getting right.
`E24_REQUIRE_GPU`	a GPU job silently writing a CPU-only cost table.
`registry.py --check`	nothing, because it was crashing: an id absent from `MAIN_CSV` made `d / ""` the *directory*, which `exists()`, so the gate had been silently covering none of E22–E29. Now 87 frozen files, 0 problems.

The deployment had a related trap: `git archive HEAD` was shipping 1.5 GB of *fetched cluster results* back to the cluster (`outputs/hpc_E22c/shapes.csv` alone is 1.28 GB). The `scp` died twice over the VPN, leaving truncated archives with no extracted tree and `.last_deploy.env` unwritten — so the E25 resubmission that was supposed to follow *silently never happened*. `export-ignore` for `outputs/hpc_*` took the archive to 94.5 MB and the deploy to three minutes.

4 Errors found and corrected

Listed because the reasoning that produced them is the part worth checking.

N-23	All four E25 jobs void: the training set was one chain replicated M times (∼30 CPU-hours). The numbers flagged themselves before the cause was known — both arms flat in M, `em_bp` worse than the project's own LMMSE floor, checkpoint pinned at the first rung, 63/64 censored.
N-23 addendum	The archival capacity sweep measured rather than inferred: 32/32 cells have exactly one unique training chain, and the M-step sees 31 distinct transitions at $M = 128$ and the same 31 at 512 — so the training-size axis was a no-op ($+0.011 \pm 0.094$ in test log-evidence over 80 paired cells). Claim E5 was already retired; this closes it with numbers.
N-24/25	The MDN arm never trained: three defects in shared EM machinery (see §2.5).
N-26	E28 void — and *my first diagnosis of it was also wrong.* The gate reported a 2.9% disagreement; I attributed it to float32 reduction-order drift from reloading 15 Sep checkpoints, on the evidence that the scatter was symmetric (111 of 192 high, 81 low) and ordered by architecture complexity (window 0.24% median, bigru 2.88% worst). That story fit the evidence and was false. The cause was that E28 hardcoded $n_{\text{test}} = 1024$ where E02a and E03 both scored on 2048; the tell was in the numbers all along — `sq_ref` 704 579 against 1 417 692, a factor 2.012. At the correct n_{test} the gate passes at 4.65×10^{-11}.
Mine, E22	I first built the ansatz from a convolution lookup applied to an already-dilated Gaussian, which measured the gap between two discretisations of the same convolution rather than model error (5.9e-8 where the ansatz is analytically exact). Rebuilt on the recursion's own kernel: 3.8e-17.
Mine, E24	Predeclared N_g exponent 2.0, measured 1.27/1.47. The `anatomy` part, added in response, is what redirected E23 away from FFT acceleration.
Mine, E23	I expected Gauss–Legendre quadrature to be an obvious improvement. It is ∼10× worse.
Mine, E27	I predicted one exact sweep would beat the 400-step reverse SDE. It ties; both are at the Monte Carlo floor. This corrected E26's reading rather than E27's.
Mine, E29	I predicted risk falls monotonically with parameter count. At small M it rises.
Mine, Q5	Two explanations for the cumulant failure refuted before the third stood, and the first Cramér test was wrong in both its variable and its refinement (§1.3).
Mine, Q1	The lemma drafted for the Kalman identification was false as stated, and its novelty claim was false too (§1.1).

5 What is running

Nineteen jobs on the Bocconi cluster at the time of writing.

jobs	what	why it matters
671442–5	E25, third attempt	**Q6**, the one open question: the neural kernel with the N-24/N-25 fixes.
671661–4	E25 at matched EM budget	removes the 1200-against-400 iteration asymmetry, so "`em_bp` had 3× the compute" cannot be said.
671654–60	E29, six architecture ladders	the arm that can refute parameter efficiency; transformer has landed, five to go.
672153	E28, resubmitted	**Q9**'s per-level ranking, now on the right bundle.
671665	E27 cost at batch 2048–32 768	states the one-sweep speed-up where BP is launch-bound and cheapest.
667192	E12b (pre-existing)	sample-complexity ladder at 16 seeds.

6 What is left to do

1. **Q6 is the only unanswered question, and it is the headline.** Read the third E25 attempt against its predeclared interpretation rule (§2.5) rather than at face value, and use the matched-budget rerun to settle the confound rather than arguing about it.

2. **Finish E29.** Five architectures outstanding. The transformer result — no capacity matches a 25-parameter mixture kernel — is strong enough that it should be attacked by the remaining five before it is believed.

3. **Q9's crossover.** The E28 smoke shows `em_bp` first at the seven lowest noise levels and the LMMSE *oracle* first at the five highest, crossing between $t = 0.4665$ and 0.6769, while the pooled scalars (0.054 against 0.163) hide it entirely. If the confirmatory run holds, the claim to make is "EM–BP dominates below t_* and the covariance twin suffices above it; we locate t_*", with the caveat that `lmmse` uses the *true* ρ and is therefore an upper bound on any learned Gaussian score. Note this is *not* `theory.crossover_time` = 1.1667, which is a different quantity.

4. **Freeze what has landed.** `icml_fetch.sh`, `registry.py --check`, `freeze.py` for E22c, E23, E24gpu2, E26, E27; `outputs/hpc_E28_VOID_estimand/` is retained deliberately, because a wrong diagnosis is only checkable against the numbers that produced it.

5. **One sentence in Proposition 1's remark** naming the Kalman smoother (§1.1). Deferred on instruction: results first, no paper work.

6. **Not planned, and deliberately so:** a Fourier BP solver (two independent measured reasons), a cumulant/Hermite solver (Cramér fails), and an FFT acceleration of the grid recursion (caps at 1.3×).

Sources: `docs/JEROME_CALL_2026-09-25.md` (the question-by-question ledger); `docs/FOURIER_ANALYSIS.md` (§1.2–1.3 in full); `docs/NEGATIVE_RESULTS.md` (N-23 to N-26); `docs/HPC_CONFIRMATORY_SET.md` (the job table); and `experiments/registry/E2*.json` for each experiment's predeclared predictions, written before its confirmatory run.